

RS001 Week 9 — English Worksheet

Topic: Cause and Effect — Enemies Appear · **Course:** Text Adventure (Python) · **Time:** about 45 minutes

This week is about thinking through how one choice changes what happens next. `if` / `else` lets different code run depending on the player's path — so the **left** road meets a goblin and the **right** road meets an orc. You will read branching code, fix broken code on paper, and explain in your own words how a choice causes an effect.

 Words to know: **if**, **else**, **elif**, **branch**, **cause**, **effect**

```
path = game.ask("Which way? (left/right)").strip().lower()
if path == "left":
    enemy = "goblin"
else:
    enemy = "orc"
game.say(f"A {enemy} appears!")
```

1 · Predict

Read each piece of code. In your head, work out what happens, then write your answer.

```
path = "left"
if path == "left":
    game.say("cave")
else:
    game.say("field")
```

Which word is printed?


```
paths = ["left", "right"]
enemies = ["goblin", "orc"]
path = "right"
if path == paths[0]:
    game.say(enemies[0])
elif path == paths[1]:
    game.say(enemies[1])
```

Which enemy is printed, and which branch ran?


```
path = "up"
if path == "left":
    game.say("goblin")
elif path == "right":
    game.say("orc")
```

The player typed **"up"** . What appears on screen?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, fix it on paper, then explain why the original was wrong.

Bug A — Left should give a goblin and right should give an orc. Right now **both** paths give a goblin.

```
path = game.ask("way?").strip().lower()
if path == "left":
    enemy = "goblin"
if path == "right":
    enemy = "goblin"
game.say(f"A {enemy} appears!")
```

Hint: the second branch has the wrong enemy.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The second branch should be an `elif`, not a fresh `if`, so the two choices are linked as one decision.

```
path = game.ask("way?").strip().lower()
if path == "left":
    game.say("A goblin blocks the cave.")
else path == "right":
    game.say("An orc roams the field.")
```

Hint: the word after the first branch should be `elif`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The enemy's name and HP should come from the **same** index. Here the name is from the left path but the HP is from the right.

```
paths = ["left", "right"]
enemies = ["goblin", "orc"]
enemy_hp = [10, 25]
path = "left"
if path == paths[0]:
    game.say(f"{enemies[0]} (HP {enemy_hp[1]})")
```

Hint: match the index across all three lists.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working example. The player chooses a path, and the choice decides which enemy appears.

```
path = game.ask("Which way? (left/right)").strip().lower()
if path == "left":
    enemy = "goblin"
    game.say("A goblin blocks the cave.")
elif path == "right":
    enemy = "orc"
    game.say("An orc roams the field.")
else:
    enemy = "shadow"
    game.say("A shadow waits on an unknown road.")
game.say(f"A {enemy} appears!")
```

What does `.strip().lower()` do to the player's answer, and why is that helpful here?

Which branch runs if the player types `right` ? Which line of `enemy =` runs?

When does the `else` branch run?

The choice is the cause. What is the effect of choosing `left` ?

Why does the last `game.say(...)` line run no matter which path the player picks?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote branching code for your own game. Make a short video on your phone (or a parent's phone) explaining the code you wrote. You may show it running. Teach it like the viewer has never coded, and use these words: **if**, **else**, **branch**, **cause**, **effect**.

1. Show the `if` / `else` code you wrote and read it out loud.
2. Explain which branch runs when the player goes left, and which runs when they go right.
3. Point out the cause (the choice) and the effect (the enemy that appears).
4. Show it running once for each path so the effect is clear.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.